

RESEARCH

Open Access



Effectiveness of a smart management system in improving adherence and clinical outcomes of patients receiving peritoneal dialysis: a retrospective cohort analysis

Yilin Zeng^{1*†}, Yan Yin^{1†}, Jia Deng¹, Dai Chen¹, Lili Deng¹, Dan Wang¹, Yujun Huang¹, Jingjun Peng^{1†} and Zhiming Ye¹

Abstract

Background Peritoneal dialysis (PD) is a widely used form of home dialysis. However, the efficacy of telehealth care in long-term management for PD patients remains unclear.

Objective The objective of this study is to assess the effectiveness of a smart management system in improving adherence, quality of life, and clinical outcomes in PD patients.

Methods The study was a retrospective cohort analysis on PD patients who underwent new catheter placement between January 2022 and June 2023. The smart PD management system, an artificial intelligence (AI)-driven, nurse-led, patient-centered management system, was used to remotely monitor patients' health at home and assist healthcare team in patient management, serving as a complement to conventional clinic visits. Patient data, including dialysis diary, symptoms, laboratory test results, and medication records, were collected through a smartphone app and uploaded to a cloud server. An AI system conducted real-time data analysis and promptly notified the healthcare team upon detecting any abnormalities. Online health coaches and nurses provided personalized guidance and interventions to patients, addressing issues in real-time. The primary endpoints of the study were changes and achievement rates in blood pressure and laboratory test results associated with complications. Secondary endpoints included changes in adherence and quality of life at recruitment and end of follow-up.

Results A total of 127 patients participated in the smart management system from January 2022 to June 2023 were included in the analysis. Most of them (94.5%) were receiving continuous ambulatory PD (CAPD). Over an average follow-up period of 11.0 (IQR 5.0–16.0) months, the average systolic blood pressure of the patients decreased from a baseline of 132.1 (SD 13.2) mmHg to 122.6 (SD 14.1) mmHg ($P < 0.001$). The overall blood pressure achievement rate ($< 140/90$ mmHg) increased from 61.4% at baseline to 79.5% ($P = 0.001$) at the end of the follow-up period. Regarding anemia, the average hemoglobin and the achievement rate increased from 98.8 (SD 19.8) g/L at baseline to 114.4 (SD 18.4) g/L ($P < 0.001$), and from 50.0 to 78.4% ($P < 0.001$), respectively. Improvements of varying degrees were observed

[†]Yilin Zeng and Yan Yin contributed equally to this work.

*Correspondence:

Yilin Zeng
498104587@qq.com

Full list of author information is available at the end of the article



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

in mineral and bone disorder, electrolyte balance, and nutritional status, even though they did not reach statistical significance. The patients' adherence to uploading dialysis diaries significantly increased from 78 to 95.7% ($P < 0.001$) during the last week of follow-up period compared to the first week after enrollment. Furthermore, there were observed improvements in the quality of life, including diet, sleep, and daily activities.

Conclusions The utilization of the smart PD management system was associated with enhanced patient adherence, improved quality of life, and superior clinical outcomes in PD patients.

Keywords Peritoneal dialysis, Self-management, End-stage kidney disease, Mobile apps, Artificial intelligence, Telehealth care, Treatment adherence, Quality of life, Patient care

Introduction

Peritoneal dialysis (PD) is one of the primary treatments for end stage kidney disease (ESKD), offering advantages such as cost-effectiveness, convenience, and better preservation of residual kidney function [1, 2]. In recent years, numerous advancements have been made in PD technology and treatment approaches. With the advancement of technology and the backing of policy, PD is expected to develop further [3–5]. By the end of 2021, approximately 126,000 ESKD patients in China were undergoing PD [6]. Upon discharge, patients are required to continue long-term PD treatment. Considering that PD is a home-based treatment method, patients and caregivers are responsible for over 90% of the treatment and care tasks. A lack of sufficient self-management skills can lead to a series of complications during home-based treatment [7] which can affect dialysis quality, increase economic and psychological stress, and potentially lead to the discontinuation of PD [8]. Therefore, it's essential to provide professional, comprehensive training and continuous follow-ups for PD patients and their caregivers.

Numerous explorations have been initiated in these fields, achieving a certain degree of success. Increasingly, PD centers are beginning to adopt digital PD management systems [9, 10]. Since 2021, we have utilized a smart PD management system at Guangdong Provincial People's Hospital in China. This system offers real-time monitoring of a patient's daily dialysis condition and promptly notifies healthcare providers of any abnormal conditions that may arise during the patient's home-based PD treatment, enabling earlier interventions. This study aims to evaluate the efficacy of the smart PD management system in improving PD treatment adherence, clinical outcomes, and quality of life in PD patients.

Methods

Study design and population

This was a single-center retrospective cohort study. Patients who initiated PD at the Peritoneal Dialysis Center of Guangdong Provincial People's Hospital between January 2022 and June 2023 and joined the smart PD management system were included in this study. Adults Patients (age ≥ 18 years old) who were undergoing regular

PD treatment, had complete baseline and follow-up data and were able and willing to provide informed consent were included. Patients were excluded if they lost to follow-up with one month of enrollment in the smart PD management system, were unable to use a smartphone, and discontinued PD treatment during follow-up (including transferred to other PD centers, underwent kidney transplantation, switched to hemodialysis, and died).

The changes in patient compliance, quality of life, and laboratory test results before and after the implementation of the smart PD management system were compared to assess its potential benefits.

Patients were included if they (1) were undergoing regular PD treatment; (2) were over 18 years old; (3) had complete baseline and follow-up data; and (4) were able and willing to provide informed consent. Patients were excluded if they (1) discontinued PD treatment during the follow-up period; (2) were under management for less than one month; or (3) were unable to use a smartphone; (4) patients discontinued PD treatment during follow-up (including transferred to other PD centers, underwent kidney transplantation, switched to hemodialysis, and died).

Patient management system and data collection

The smart PD management system was an artificial intelligence (AI)-driven, nurse-led, patient-centered collaborative management system designed to complement regular clinic visits for PD patients. This system was driven by AI, supported by a team of experienced PD nurses, and complemented by a health coach team consisting of nurses trained by nephrologists, dietitians, and social workers. Nurses and nephrologists are able to follow up on patients' home PD treatments in a timely manner, and provide professional guidance on the adjustment of treatment prescriptions based on patients' feedback on treatment problems and early warning information. At the same time, nurses and nephrologists will provide guidance on the observation of home treatment complications and self-care education to ensure that patients communicate with and consult with the hospital's professional doctors and nurses through the smart platform at any time during their home treatments. The system

included a smartphone app for patients, dashboards for health coaches and nurses, and an AI platform for data analysis (Fig. 1).

Patients who participated in the smart management system could record daily home dialysis diary, blood pressure, laboratory test results, appetite, mental status, and sleep quality through photographs (Fig. 2–1). These data were encrypted and transmitted to a cloud server. For image data, information was extracted from the images using a deep-learning optical character recognition system, and then structured into a database following manual verification. For patients undergoing automated peritoneal dialysis (APD), real-time data, including dialysis mode and ultrafiltration volume, were uploaded to the server via Bluetooth, Wi-Fi, and other communication protocols. The AI engine on the server performed real-time analysis of the data, and the system would send alerts to the healthcare team if the abnormal situations necessitating intervention were detected. The online health coaches communicated with patients through the patient-end app (Fig. 2–2), and provided initial intervention. If necessary, the patients would be referred to PD center for further assessment and treatment. In addition to providing real-time responses to patients' inquiries, the online multidisciplinary health coach team and medical staff also proactively engaged in educating and intervening in patients' diet, exercise, sleep, and psychological well-being. The intelligent nutrition prescription system we had developed could generate personalized meal plans in accordance with the nutritional guidelines, considering the patient's disease status and food preferences.

Meanwhile, patients could record their daily meals using the built-in tools within the app and check whether their daily nutrient intake aligned with their individual requirements.

Outcomes

The primary endpoints of this study were the changes and achievement rates in patients' blood pressure and laboratory test results before and after the implementation of the management system. (hemoglobin, serum albumin, potassium, calcium, phosphorus, and intact parathyroid hormone (iPTH)) the achievement rates for blood pressure and laboratory test results were as follows: ①Blood pressure: <140/90 mmHg; ②Hemoglobin: >100 g/L; ③Serum potassium: 3.5–5 mmol/L; ④Serum calcium: 2.1–2.5 mmol/L; ⑤Serum phosphorus: 1.13–1.78 mmol/L; ⑥iPTH: 2–9 times the upper limit of the normal reference range; ⑦ Serum albumin: >35 g/L.

The secondary endpoints were the adherence to dialysis diary (adherence to dialysis diary is defined as peritoneal dialysis patients voluntarily uploading complete data to the APP greater than 5 days per week) and changes in patient-reported quality of life which show by three-dimensional questionnaire that included appetite, mental status, and sleep quality. A three-dimensional questionnaire including appetite, mental status and sleep quality was used to find out the changes in patient-reported quality of life. This questionnaire was based on the SF-36 questionnaire and was selected to evaluate the quality of life of the peritoneal dialysis patients in terms of appropriate dimensions, including appetite, mental and sleep

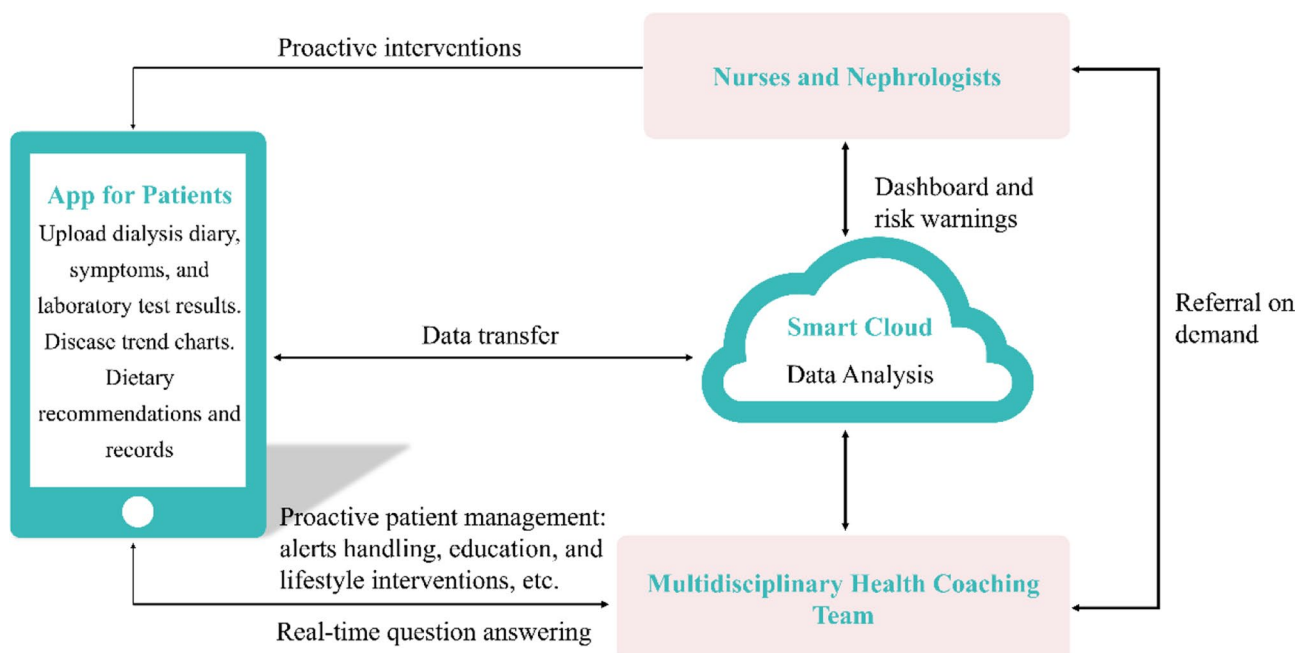


Fig. 1 Framework of smart PD management system

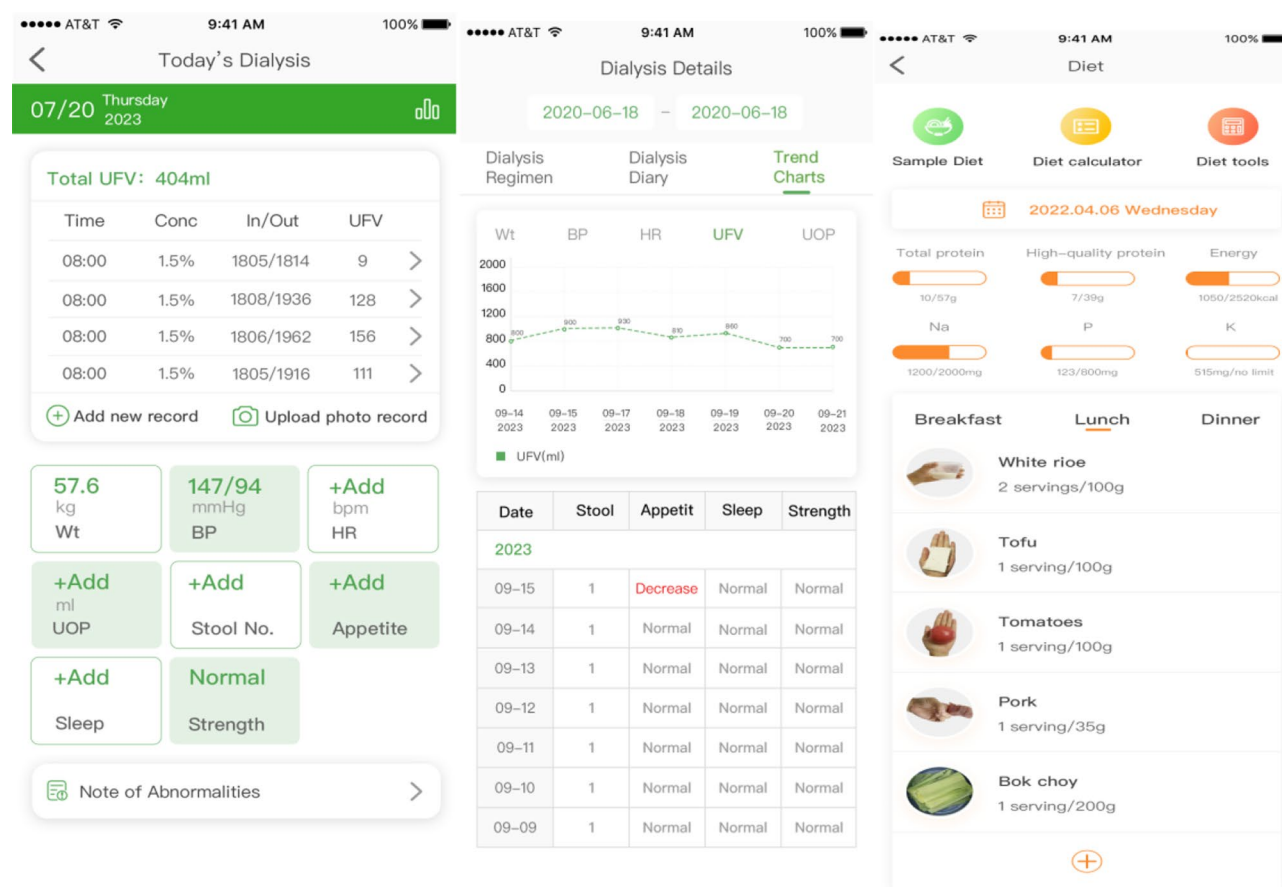


Fig. 2–1 Recording of daily home dialysis diary, dietary diary

quality. The assessment in these three dimensions was scored (worse compared to before = 0; no change compared to before = 1; and getting better compared to before = 2, with a total score of 6.) The higher the score, the better the patient's state of life. Good adherence was defined as timely uploading of the dialysis diary for a minimum of 5 days per week. According to clinical guidelines [11–14] and local laboratory reference ranges and the questionnaire form is as follows:

Content	Options	Score
appetite	worse compared to before	0
	no change compared to before	1
	getting better than before	2
mental	worse compared to before	0
	no change compared to before	1
	getting better than before	2
sleep quality	worse compared to before	0
	no change compared to before	1
	getting better than before	2
Total Score		

Statistical analyses

The distribution of continuous variables was described by the mean (standard deviation, SD) or median (inter-quartile range, IQR). The comparison of mean values of clinical indicators before and after management was conducted using paired t-test or Wilcoxon signed-rank test. Meanwhile, the comparison of achievement rates of indicators (expressed as percentages) before and after management was conducted using McNemar's test. The statistical analysis was performed using R (version 4.1.2), and a two-sided test with a P value < 0.05 was considered statistically significant.

Results

Baseline characteristics

From January 2022 to June 2023, a total of 145 PD patients participated in the smart PD management system. Among them, 18 (12.4%) patients discontinued PD treatment during follow-up (including 11 who transferred to other PD centers, 3 who underwent kidney transplantation, 1 who switched to hemodialysis, and 3 who died). 127 patients met the inclusion and exclusion criteria, the follow-up of this patients start at the first day they begin the PD treatment including daily online

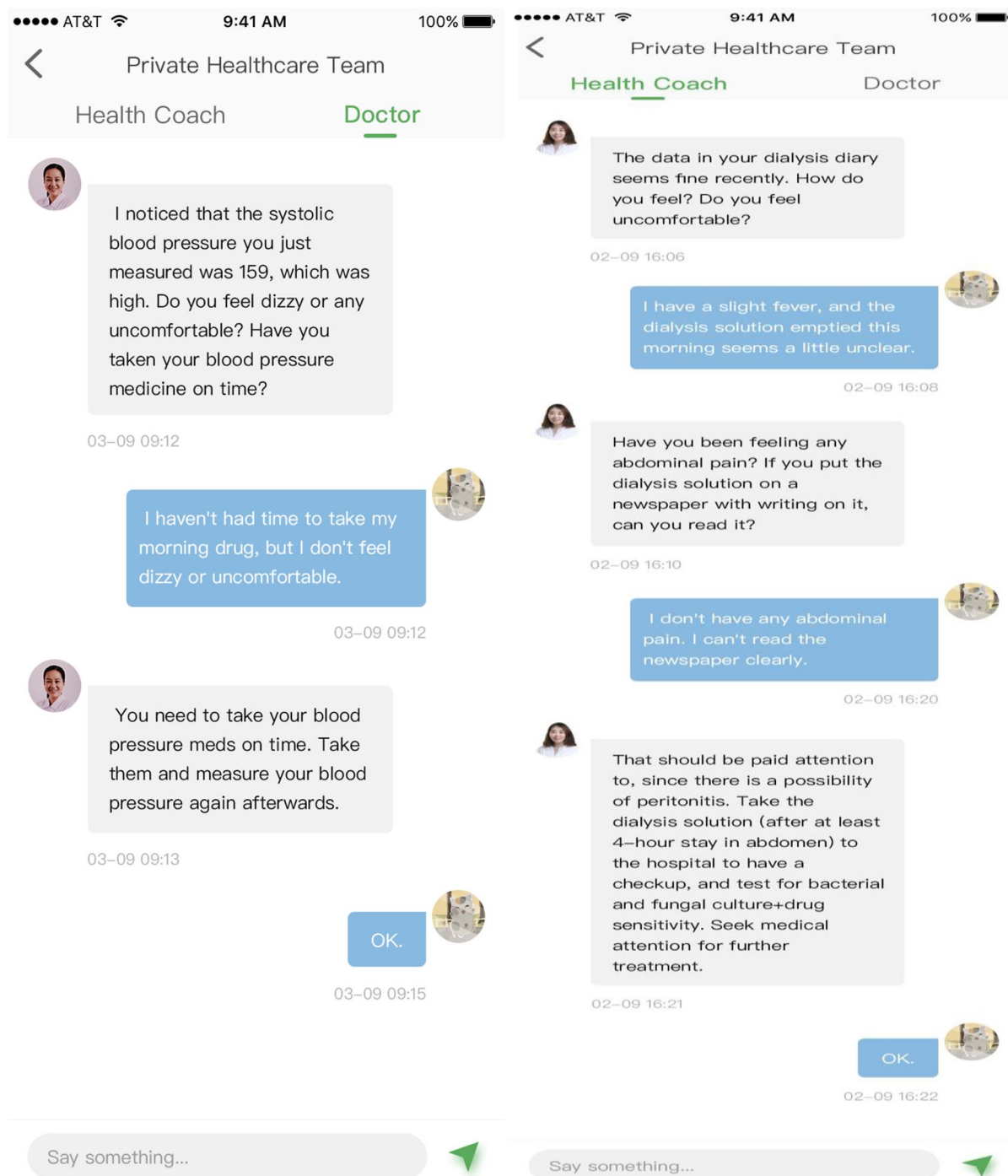


Fig. 2–2 Online health coaches communicating with patients in the patient-end app

feedback on home dialysis using a smart program and regular monthly visits to outpatient clinics for home treatment assessment, of whom 59 (46.5%) were female. The youngest patient was 19 years old, the oldest was 68 years old, and the average age was 47.2(SD 13.0) years old. The baseline average BMI was 22.5 (SD 3.8) kg/m². 120 (94.5.0%) patients received continuous ambulatory PD (CAPD). The primary disease in 52 (40.9%) patients

was primary chronic glomerulonephritis. 109 (85.8%) patients had a urine output of >400 ml/24 h. The average dialysis duration was 11.0 (IQR 5.0–16.0) months (Table 1).

Changes in blood pressure

At the end of the follow-up period, the average systolic blood pressure of patients decreased from a baseline

Table 1 Baseline characteristics

Indicators	Value
Age (years)	47.2(SD 13.0)
Sex	
Female	59(46.5%)
Male	68(53.5%)
Baseline BMI (kg/m²)	22.5(SD 3.8)
PD modality	
CAPD	120(94.5%)
APD	1(0.8%)
IPD	3(2.4%)
CAPD + APD	3(2.4%)
Etiology	
Primary chronic glomerulonephritis	52 (40.9%)
Diabetic nephropathy	6(4.7%)
Hypertensive nephropathy	6(4.7%)
Unknown or other kidney diseases	63(49.6%)
Urine output (ml/24h)	
> 400	109(85.8%)
100–400	16(12.6%)
< 100	2(1.6%)
Dialysis duration (month)	11.0(IQR 5.0–16.0)
Baseline blood pressure (mmHg)	
Systolic pressure	132.1(SD 13.2)
Diastolic pressure	82.7(SD 9.9)
CAPD: continuous ambulatory peritoneal dialysis	
IPD: intermittent peritoneal dialysis	
CCPD: continuous cycling peritoneal dialysis	
APD: automated peritoneal dialysis	

Table 2 Changes in laboratory test results

Indicators	Baseline	End of follow-up	P value
Hemoglobin			
Value	98.8(SD 19.8)	114.4(SD 18.4)	< 0.001
Achievement rate	50.0%	78.4%	< 0.001
Serum albumin			
Value	35.1(SD 4.8)	35.5(SD 3.8)	0.596
Achievement rate	56.7%	59.6%	0.755
Calcium			
Value	2.21(SD 0.26)	2.27(SD 0.19)	0.005
Achievement rate	62.6%	70.4%	0.243
Phosphorus			
Value	1.73(SD 0.49)	1.70(SD 0.52)	0.254
Achievement rate	50.0%	50.1%	1.000
iPTH			
Value	237.3(SD 174.2)	261.8(SD 201.8)	0.134
Achievement rate	68.2%	72.7%	0.774
Potassium			
Value	4.0(SD 0.7)	3.9(SD 0.6)	0.527
Achievement rate	54.8%	66.1%	0.098

measurement of 132.1 (SD 13.2) mmHg to 122.6 (SD 14.1) mmHg ($P < 0.001$), with an average decrease of 9.5 mmHg. The diastolic pressure decreased from the baseline value of 82.7 (SD 9.9) mmHg to 79.7 (SD 9.7)

mmHg ($P = 0.003$), with an average decrease of 3 mmHg. The overall achievement rate for blood pressure control ($< 140/90$ mmHg) increased from 61.4% at baseline to 79.5% ($P = 0.001$).

Changes in laboratory results

In terms of anemia, the average hemoglobin level and the achievement rate increased from 98.8 (SD 19.8) g/L at baseline to 114.4 (SD 18.4) g/L ($P < 0.001$), and from 50.0 to 78.4% ($P < 0.001$), respectively. As for mineral and bone disorder and electrolytes, although not statistically significant, the achievement rates for serum calcium, phosphorus, iPTH, and potassium increased from 62.6%, 50.0%, 68.2%, and 54.8% at baseline to 70.4%, 50.1%, 72.7%, and 66.1%, respectively. Regarding nutritional status, the achievement rate for serum albumin increased from 56.7 to 59.6% (Table 2).

Improvement in adherence

Defined by the requirement of timely uploading of the dialysis diary for a minimum of 5 days per week. uploading dialysis diaries on time for ≥ 5 days per week, patient adherence to uploading dialysis diaries significantly increased from 78 to 95.7% ($P < 0.001$) during the last week of follow-up period, compared to the first week after enrollment. In terms of the number of data uploads, the average monthly number of data uploads per person for dialysis diaries, blood pressure, bowel movements, and weight were 21.6, 28.1, 20.5, and 23.5 respectively during the first month of enrollment. They increased to 25.9, 34.1, 23.6, and 26.6 respectively in the 3rd month of follow-up, and further increased to 27.8, 38.4, 26.1, and 26.9 respectively in the 12th month of follow-up (Fig. 3).

Changes in appetite, mental status, and sleep quality

By using a three-dimensional questionnaire that included appetite, mental status, and sleep quality as self-reported by the patients to understand how patients adapt to life in home treatment. The patients were instructed to adhere to daily evaluation if possible, with the scoring protocol defined as follows: good = 2 points, fair = 1 point, and poor = 0 points. The results showed that the average scores for appetite, mental state, and sleep quality during the first month of enrollment were 1.41 (SD 0.43), 1.44 (SD 0.42), and 1.30 (SD 0.45) respectively. These scores increased to 1.50 (SD 0.49, $P = 0.163$), 1.51 (SD 0.49, $P = 0.242$), and 1.35 (SD 0.51, $P = 0.464$) respectively during the sixth month, and further increased to 1.60 (SD 0.53, $P = 0.028$), 1.61 (SD 0.53, $P = 0.045$), and 1.40 (SD 0.52, $P = 0.176$) respectively during the twelfth month (Fig. 4).

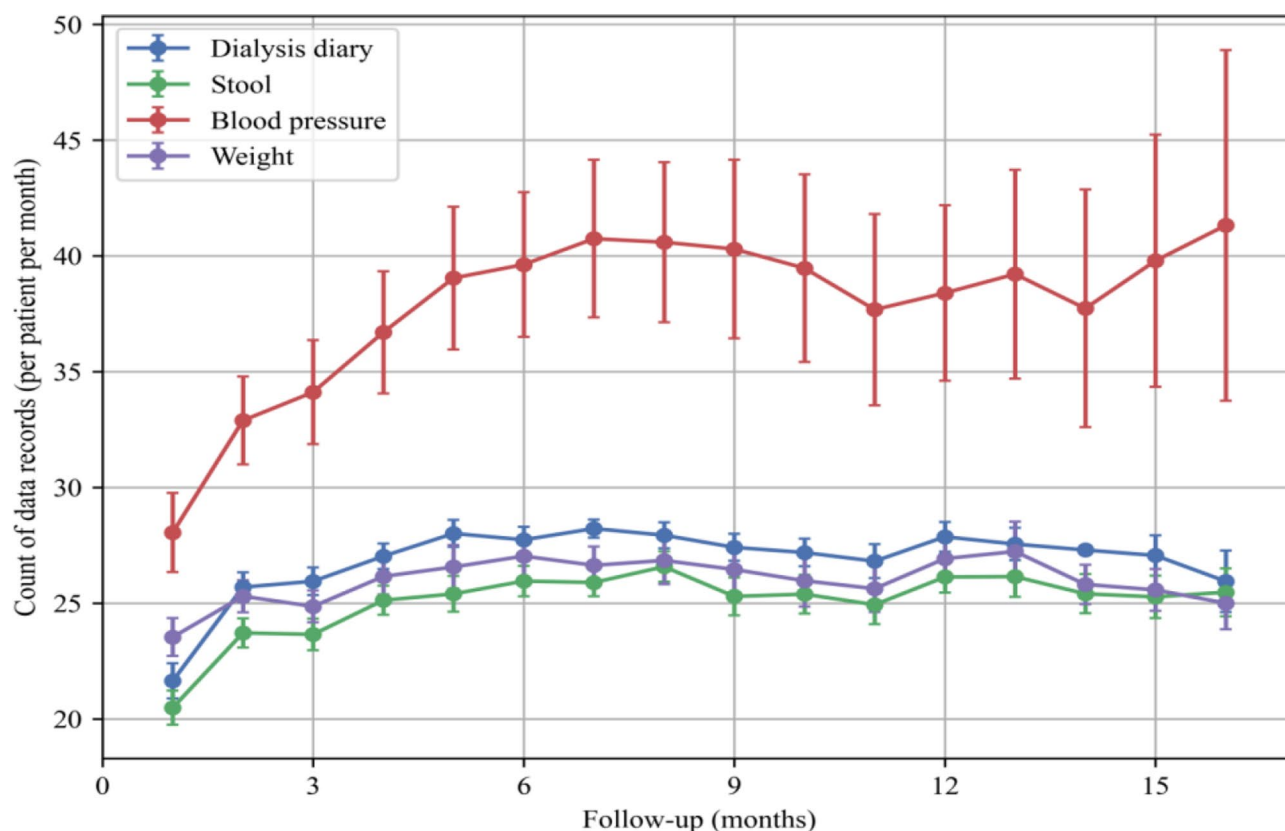


Fig. 3 The trend of adherence to data uploading

Discussion

This study found that the application of a smart PD management system was associated with patients' clinical benefits, particularly in terms of blood pressure reduction and anemia improvement. The reasons for these effects were multifaceted, with one important factor being the improvement in patient treatment compliance. This improvement was achieved through the utilization of the AI system and the sustained proactive interventions of a multidisciplinary team in areas such as patient education, dietary guidance, and lifestyle interventions. Consequently, patients' self-management abilities and treatment compliance were strengthened, resulting in improved dialysis quality and a reduction in complications. On the other hand, real-time alerts and online consultations were utilized to promptly intervene in abnormal situations, reducing the risk of potential serious adverse events.

As a home-based kidney replacement therapy, PD is relatively simple in technique and can better protect residual kidney function. It has several advantages including reduced reliance on treatment facilities and equipment, relatively low costs, ease of operation, and helping patients reintegrate into society. In recent years, PD has been widely used among ESKD patients worldwide, especially in developing countries [2, 15]. However,

the management of PD treatment still faces challenges and concerns, particularly in terms of patient education and long-term care, which can lead to poor compliance and adverse clinical outcomes. Studies have shown that nearly 30% of PD patients could not adhere to the dialysis prescription, which may result in adverse clinical outcomes and increased treatment costs [16–18]. Since patients usually receive home-based treatment and only visit doctor at hospital every 1–2 months, some PD-related complications and adverse events could not be easily detected early on. Even if patients can quickly identify issues related to PD, the time and transportation costs required to go to the hospital may hinder them from receiving appropriate treatment.

In view of this, the use of telehealth system for remote monitoring and management of PD patients is expected to be a promising solution [19–21]. It is noteworthy that our recent research [22] had revealed that the use of a similar system, known as KidneyOnline, was associated with reduced risk of unfavorable kidney outcomes in patients with chronic kidney disease. In terms of PD, a small-scale observational study [23] showed that after 6 months of remote monitoring and management of PD patients using automated peritoneal dialysis (APD), patient compliance, blood pressure, and dialysis adequacy all improved to varying degrees. Another multicenter

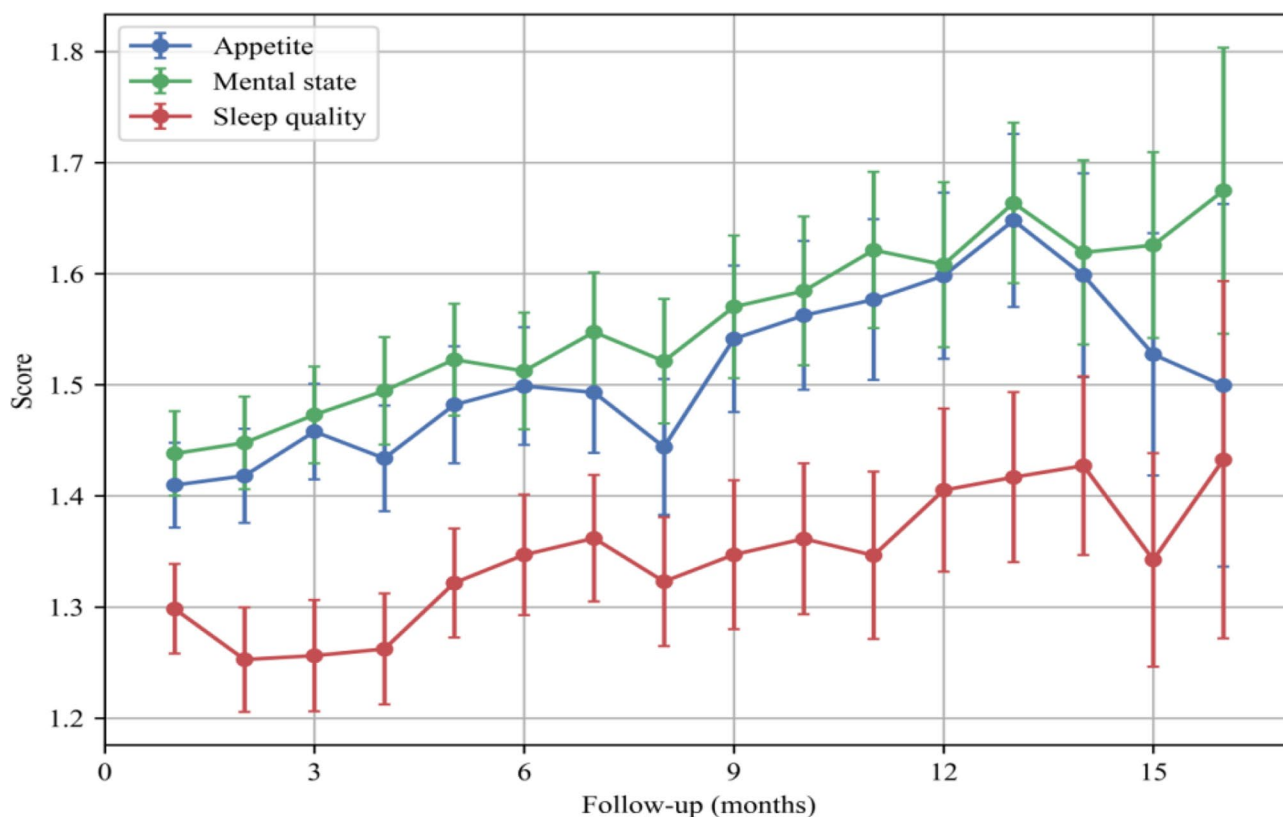


Fig. 4 The trend of appetite, mental status, and sleep quality

randomized controlled study [24] found that remote patient management could improve the satisfaction of PD patients but has no significant impact on quality of life. However, existing studies [25–28] are exploratory, with relatively limited monitoring and interventions, short follow-up periods, and small sample sizes. Moreover, they mostly used patient satisfaction, quality of life, or blood pressure as endpoints. There is still a lack of long-term follow-up to comprehensively assess the benefits of a smart PD management system on patient compliance and clinical outcomes.

Currently, the majority of hospitals in China are still using traditional face-to-face follow-up (95.6%) for local PD patients and telephone follow-up (91.1%) for PD patients from other regions. Furthermore, it is worth noticing that 66.5% of these hospitals do not possess a digital PD patient management system [2, 29]. The smart PD management system we developed utilized smartphone technology, the Internet of Things, and AI technologies to address the pain points of traditional PD management, such as inadequate home management, education, and follow-up. It expanded the scope of PD management from the hospital to the patient's home, forming an integration of in-hospital and out-of-hospital care, as well as a fusion of offline and online services. This

comprehensive management loop provided effective support for patients' home-based PD treatment.

To the best of our knowledge, this study is the first to evaluate the effect of an AI-driven, smartphone-based, nurse-led, patient-centered management system on improving adherence and clinical outcomes among PD patients, particularly those undergoing CAPD. However, our study has certain limitations. Firstly, our study was designed as a single-center retrospective study, which presents challenges in establishing causality and introduces the potential for selection bias. Additionally, our approach involved within-subject comparisons rather than a parallel control group. Nevertheless, our findings suggested that this management system contributed to the enhancement of patient adherence, their quality of life, and clinical outcomes. Thirdly, we did not consider other potential outcomes, such as PD discontinuation or transfer to hemodialysis, cardiovascular complications, healthcare costs, and hospital readmission rates. These additional parameters could have provided a more comprehensive assessment of the effectiveness of our system. Therefore, more data is needed to offer greater insights into the benefits of this care system.

Conclusion

In this study, we developed an artificial intelligence-driven, smartphone-based, nurse-led, patient-centered home dialysis care management system. Through real-time alerts and real-time guidance, we made home peritoneal dialysis treatments safer for patients, while increasing patient compliance with constant communication and professional treatment guidance, which led to better quality of peritoneal dialysis treatments and improved patients' appetite, spirit, and appetite.

These findings provide evidence for telemedicine interventions and a new direction for home peritoneal dialysis management, however, superior to the limited sample size of this study and the short study duration of patients, we look forward to conducting prospective clinically controlled experiments with larger sample sizes in the future to collect more valid data, and we expect to discover really effective peritoneal dialysis follow-up tools and interventions to continuously improve the quality of dialysis and the high quality of the long-term management of peritoneal dialysis patients.

Abbreviations

PD	Peritoneal dialysis
ESKD	End stage kidney disease
AI	Artificial intelligence
iPTH	Intact parathyroid hormone
SD	Standard deviation
IQR	Interquartile range
CAPD	Continuous ambulatory peritoneal dialysis
APD	Automated peritoneal dialysis

Acknowledgements

Not applicable.

Author contributions

Yilin Zeng and Zhiming Ye researched literature and conceived the study. Jia Deng, Lili Deng, Jingjun Peng, Yan Yin, Dan Wang, Dai Chen, Yujun Huang were involved in protocol development, gaining ethical approval, patient recruitment and data analysis. Yilin Zeng and Yan Yin wrote the first draft of the manuscript. All authors reviewed and edited the manuscript and approved the final version of the manuscript.

Funding

(1) Guangdong Provincial Science and Technology Planning Project (Grant No.2023B110009). (2) Guangzhou Science and Technology Project (No. 202206080010).

Data availability

All data in this study were collected from patients in our center through follow-up visits. If the publisher and reviewers have any requests for data, they can contact the first author Yilin Zeng directly.

Declarations

Ethics approval and consent to participate

This study The experiment has or participants have given informed consent and signed an informed consent form and approved by the Ethics Committee of Guangdong province people's hospital (NO.KY2023-646), which operates in accordance with ethical principles such as the Declaration of Helsinki and the requirements of national laws and regulations: (1) All biomedical researches involving human beings need to be registered in the Medical Research Registration and Filing Information System (MRFIS) for record-keeping after ethical approvals have been granted. All biomedical research involving human

beings must be registered in the "Medical Research Registration and Filing Information System" after ethical approval before launching. (2) In order to fully guarantee the rights and interests of the research subjects, if there are any changes in the research protocol, informed consent and recruitment materials, or if there are any serious adverse events (in human) or other unforeseen circumstances after the commencement of the trial, please report them to the Committee, which has the right to decide whether or not to terminate the trial. (3) For projects with a duration of more than one year, an annual progress report shall be submitted to the Ethics Committee for follow-up and review. (4) submit the research project completion report to the Ethics Committee for review.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Guangdong Provincial People's Hospital, Guangzhou, China

Received: 28 October 2024 / Accepted: 27 June 2025

Published online: 07 July 2025

References

1. Xiao D, Haishan W, Xueqing Y, Xiao Y. Research progress of technique failure and its risk factors in early peritoneal dialysis. *Chin J Nephrol.* 2022;38:144–50. <https://doi.org/10.3760/cmaj.cn441217-20210721-00022>.
2. Project group of. White paper on the current status of peritoneal dialysis management in China. *Chin J Nephrol.* 2022;38:1076–104. <https://doi.org/10.3760/cmaj.cn441217-20220418-00158>.
3. Mehrotra R, Devuyst O, Davies SJ, Johnson DW. The current state of peritoneal Dialysis. *J Am Soc Nephrol.* 2016;27:3238–52. <https://doi.org/10.1681/AS.N.2016010112>.
4. Li PK, Chow KM, Van de Luijngaarden MW, Johnson DW, Jager KJ, Mehrotra R, Naicker S, Pecoits-Filho R, Yu XQ, Lameire N. Changes in the worldwide epidemiology of peritoneal dialysis. *Nat Rev Nephrol.* 2017;13:90–103. <https://doi.org/10.1038/nrneph.2016.181>.
5. Kim H, Ryu DR. A prime determinant in selecting dialysis modality: peritoneal dialysis patient survival. *Kidney Res Clin Pract.* 2017;36:22–8. <https://doi.org/10.23876/j.krcp.2017.36.1.22>.
6. Ya-nan C, Yong-ni W, Cai-ping Z, Xiao-qin M, Shu-hua S, Yan W. Visualization analysis of research hotspots in the field of peritoneal dialysis nursing based on the software of CiteSpace. *Chin J Blood Purif.* 2023;22:232–6.
7. Yu X, Mehrotra R, Yang X. Components of a successful peritoneal dialysis program. *Semin Nephrol.* 2017;37(1):10–16. <https://doi.org/10.1016/j.semnephrol.2016.10.003>. PMID: 28153189.
8. Piarulli P, Vizzardi V, Alberici F, Riva H, Aramini M, Regusci L, Cippà P, Bellasi A. Peritoneal dialysis discontinuation: to the root of the problem. *J Nephrol.* 2023;36(7):1763–76. <https://doi.org/10.1007/s40620-023-01759-w>. Epub 2023 Sep 25. PMID: 37747660; PMCID: PMC10543152.
9. Xiaoning X, Xiaochan H, Huimei L, Wei D, Xiaojun T, Haihong Z. Application of internet-based continuous nursing in home-based maintenance peritoneal dialysis. *Mod Clin Nurs.* 2021;20. <https://doi.org/10.3969/j.issn.1671-8283.2021.04.010>.
10. Yilan L, Dejjiao H, Zhongxiang C. Progress in training and follow-up of peritoneal dialysis patients. *Chin J Blood Purif.* 2021;20:269–72.
11. Chinese Society of Nephrology. Guidelines for hypertension management in patients with chronic kidney disease in China (2023). *Chin J Nephrol.* 2023;39:48–80. <https://doi.org/10.3760/cmaj.cn441217-20220630-00650>.
12. McMurray JJV, Parfrey PS, Adamson JW, Aljama P, Berns JS, Bohlius J, Drüeke TB, Finkelstein FO, Fishbane S, Ganz T, et al. Kidney disease: improving global outcomes (KDIGO) anemia work group. KDIGO clinical practice guideline for anemia in chronic kidney disease. *Kidney Int Suppl.* 2012;2:279–335. <https://doi.org/10.1038/kisup.2012.37>.
13. Levin A, Stevens PE, Bilous RW, Coresh J, Winearls CG. Kidney disease: improving global outcomes (KDIGO) CKD work group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl.* 2013;3:1–150. <https://doi.org/10.1038/kisup.2012.73>.

14. Kidney Disease: Improving Global Outcomes (KDIGO) CKD-MBD Update Work Group. KDIGO 2017 clinical practice guideline update for the diagnosis, evaluation, prevention, and treatment of chronic kidney Disease–Mineral and bone disorder (CKD-MBD). *Kidney Int Suppl.* 2017;7:1–59. <https://doi.org/10.1016/j.kisu.2017.04.001>.
15. Xuelian B, Jiayi Z, Guoliang X, Xiaoqing C. Meta-analysis on the prevalence of chronic kidney disease in Chinese adults. *China Med Pharm.* 2022;012:49–53.
16. Bernardini J, Nagy M, Piraino B. Pattern of noncompliance with dialysis exchanges in peritoneal dialysis patients. *Am J Kidney Dis.* 2000;35:1104–10. [https://doi.org/10.1016/s0272-6386\(00\)70047-3](https://doi.org/10.1016/s0272-6386(00)70047-3).
17. Gexia Z, Qian L, Yanshuang L, Jing Z, Zheng C, Xiaoyu L, Jinghua S. Application of medical failure mode and effect analysis in the management of home peritoneal dialysis patients. *Chongqing Med.* 2022;51. <https://doi.org/10.3969/j.issn.1671-8348.2022.01.038>.
18. Haiyan C. Effect of follow-up nursing management measures guided by WeChat platform on peritoneal dialysis at home. *J Clin Pathological Res.* 2021;41:866–71. <https://doi.org/10.3978/j.issn.2095-6959.2021.04.021>.
19. John O, Jha V. Remote patient management in peritoneal dialysis: an answer to an unmet clinical need. *Contrib Nephrol.* 2019;197:99–112. <https://doi.org/10.1159/000496305>.
20. Yeter HH, Manani SM, Ronco C. The utility of remote patient management in peritoneal dialysis. *Clin Kidney J.* 2021;14:2483–9. <https://doi.org/10.1093/ckj/sfab111>.
21. Lew SQ, Sikka N, Thompson C, Cherian T, Magnus M. Adoption of telehealth: remote biometric monitoring among peritoneal Dialysis patients in the united States. *Perit Dial Int.* 2017;37:576–8. <https://doi.org/10.3747/pdi.2016.0272>.
22. Liu W, Yu X, Wang J, Zhou T, Yu T, Chen X, Xie S, Han F, Wang Z. Improving kidney outcomes in patients with nondiabetic chronic kidney disease through an artificial Intelligence-Based health coaching mobile app: retrospective cohort study. *Jmir Mhealth Uhealth.* 2023;11:e45531. <https://doi.org/10.2196/45531>.
23. Yeter HH, Akcay OF, Ronco C, Derici U. Automated remote monitoring for peritoneal Dialysis and its impact on blood pressure. *Cardiorenal Med.* 2020;10:198–208. <https://doi.org/10.1159/000506699>.
24. Jung HY, Jeon Y, Kim YS, Kim DK, Lee JP, Yang CW, Ko EJ, Ryu DR, Kang SW, Park JT, et al. Outcomes of remote patient monitoring for automated peritoneal dialysis: A randomized controlled trial. *Nephron.* 2021;145:702–10. <https://doi.org/10.1159/000518364>.
25. Biebuyck G, Neradova A, de Fijter C, Jakulj L. Impact of telehealth interventions added to peritoneal dialysis-care: a systematic review. *Bmc Nephrol.* 2022;23:292. <https://doi.org/10.1186/s12882-022-02869-6>.
26. Olivares-Gandy HJ, Dominguez-Isidro S, Lopez-Dominguez E, Hernandez-Velazquez Y, Tapia-McClung H, Jorge D. A telemonitoring system for nutritional intake in patients with chronic kidney disease receiving peritoneal dialysis therapy. *Comput Biol Med.* 2019;109:1–13. <https://doi.org/10.1016/j.compbiomed.2019.04.012>.
27. Milan MS, Baretta M, Giuliani A, Virzi GM, Martino F, Crepaldi C, Ronco C. Remote monitoring in peritoneal dialysis: benefits on clinical outcomes and on quality of life. *J Nephrol.* 2020;33:1301–8. <https://doi.org/10.1007/s40620-020-00812-2>.
28. Uchiyama K, Morimoto K, Washida N, Kusahana E, Nakayama T, Itoh T, Kasai T, Wakino S, Itoh H. Effects of a remote patient monitoring system for patients on automated peritoneal dialysis: a randomized crossover controlled trial. *Int Urol Nephrol.* 2022;54:2673–81. <https://doi.org/10.1007/s11255-022-03178-5>.
29. Wenwen W, Yanpei C, Yanhong G, Xiaolin G, Chunhua H. Investigation of the status quo and intention to seek healthcare among peritoneal Dialysis patients in Shanghai tertiary general hospitals. *Shanghai Nurs J.* 2023;23:34–8. <https://doi.org/10.3969/j.issn.1009-8399.2023.02.008>.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.